

Bayesian Statistics and Applications

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Probability Primer

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- Conditional Probabilities: How we incorporate new information. For example, what's the probability a coin is fair, if it came up heads 100 times in a row?

$$\Pr[X|Y] = \frac{\Pr[X \text{ and } Y]}{\Pr[Y]}$$

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- Repeat, using the new belief this time as the initial belief next time.

Expected Value

Motivating Example

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- Expectation is taken over certain *beliefs* (what if the coin actually isn't fair?).
- We can *update* these beliefs as we go along! (using Bayes' thm. from last slide).

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- Defines a (slightly-notation-abusing) mass function on subsets. If S is the sample space, $f(s \subseteq S)$ is the probability that X takes on any value $x \in s$.
- If there's an ordering on the outcomes (usually they're numerical), defines a *cumulative mass function*, $F(x)$, which is $\Pr[X \leq x]$.

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 - Expectation: $\sum x_i \Pr[X_i]$ vs. $\int \theta f(\theta) d\theta$.
 - Variance: $\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$ in both cases.
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- Calculating probabilities for regions: integrate. [Example with uniform distribution].
- Defines cumulative density function $F(x)$, which is $\int_{-\infty}^x f(\theta) d\theta$ (how much density is below x)?

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- Say we start with a *uniform prior* about the probability of heads, θ . Why does this represent no knowledge?
- So our prior distribution is $f(\theta) = 1$ for $\theta \in [0, 1]$. Say x is the "data" we collect by this experiment, or the number of heads in three tosses. We have:

$$\Pr[x = 3|\theta] = \theta^3$$

$$\Pr[x \neq 3|\theta] = 1 - \theta^3$$

- Calculating total probability:

$$\Pr[x = 3] = \int_0^1 \Pr[x = 3|\theta] \Pr[\theta] d\theta = 0.25$$

Bayes' Rule Example, Cont'd.

So we can now use Bayes' Rule:

$$\begin{aligned}f(\theta|x=3) &= \frac{\Pr[x=3|\theta]f(\theta)d\theta}{\Pr[x=3]} \\ &= 4\theta^3d\theta\end{aligned}$$

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What if we used certain priors ($\theta = \delta_x$)?

Posterior Belief Plot

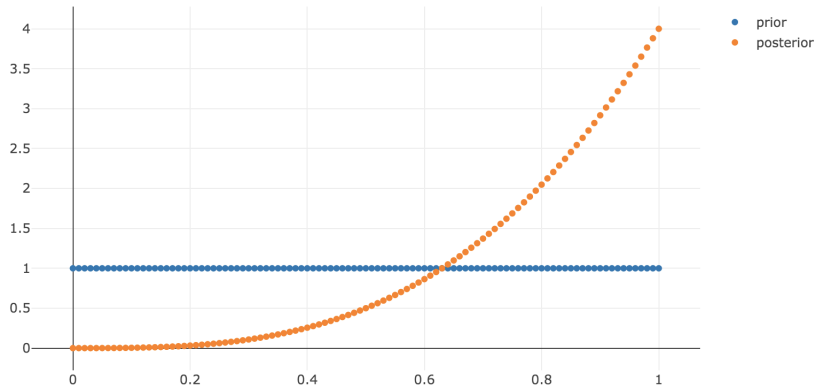


Figure: Belief Revision

Naive Bayesian Spam Filter

Motivating Example

For a given word, W , there's a certain probability that the word appears in spam ($\Pr[W|S]$), and a certain probability that a random email with the word in it is spam ($\Pr[S|W]$). The first one is observable, and the second one isn't (but we want to know it).

How might we approach this using Bayes' theorem?

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What about multiple words?

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Paul Graham's Formula:

$$\Pr[S|W_1, \dots, W_N] = \frac{\prod \Pr[S|W_i]}{\prod \Pr[S|W_i] + \prod (1 - \Pr[S|W_i])}$$

What does this assume? [Hint: Why is this kind of thing "naive?"]

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Things to consider (more data/different kinds of data, penalize false positives, how and which tokens to take).

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- For Bayesians, probabilities are **in the mind**.
- Bayes' Rule gives us a way to **update** our beliefs based on what we observe.
- Probabilities obey certain rules (like $0 \leq p \leq 1$), and have certain useful functions (like expectation $\mathbb{E}$).

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